

SC26 Figures 3-6: reproduction outcome

Environment: clariden-ln004 · fresh run output: /iopsstor/scratch/cscs/btommaso/sc26_reproduction_20260722/reproduction_clariden-ln004_20260722T013845

Figure	Paper panel	Input evidence	Fresh result in this run
3	AllReduce 128 MiB, 16N/64 GPU, ch1 + auto	exact original NSYS	raw NSYS → fresh SQLite → fresh GOAL/metadata → LGS/LP
4	Mixed20, 16N/64 GPU, ch1	two raw candidates	old job reproduces curve; corrected written-config campaign does not
5	Llama N4/GPU16/DP16	related static BIN + old metadata	paper curve not reproduced; raw trace/GOAL absent; 7B/8B conflict
6 Llama	N32/GPU128/DP128	derived metadata only	fresh Composite latency/bandwidth; paper says 70B, metadata says 7B
6 vLLM	Llama-3.1-8B, N2/GPU8, 128 tokens	original expired	not rerunnable; displayed as blocked, never substituted with 70B

Match against packaged paper/reference CSVs:

- Fig. 3 ch1: latency Monolithic 0.078% max / 0.025% mean; bandwidth Monolithic 2.373% max / 0.230% mean
- Fig. 3 auto: latency Monolithic 0.166% max / 0.056% mean; bandwidth Monolithic 1.493% max / 0.523% mean
- Fig. 3 LGS: ch1 latency 5.256% max / 0.646% mean; auto latency 2.285% max / 0.548% mean
- Fig. 4 old curve-source: Monolithic 0.118% max / 0.060% mean; LGS 2.076% max / 0.612% mean
- Fig. 4 corrected raw campaign: Composite 297.392% max / 196.042% mean; Monolithic 306.871% max / 248.451% mean
- Fig. 5 old metadata Composite vs packaged paper curve: 15.910% max / 10.038% mean
- Fig. 6 Llama metadata: latency 0.128% max / 0.057% mean; bandwidth 0.358% max / 0.026% mean

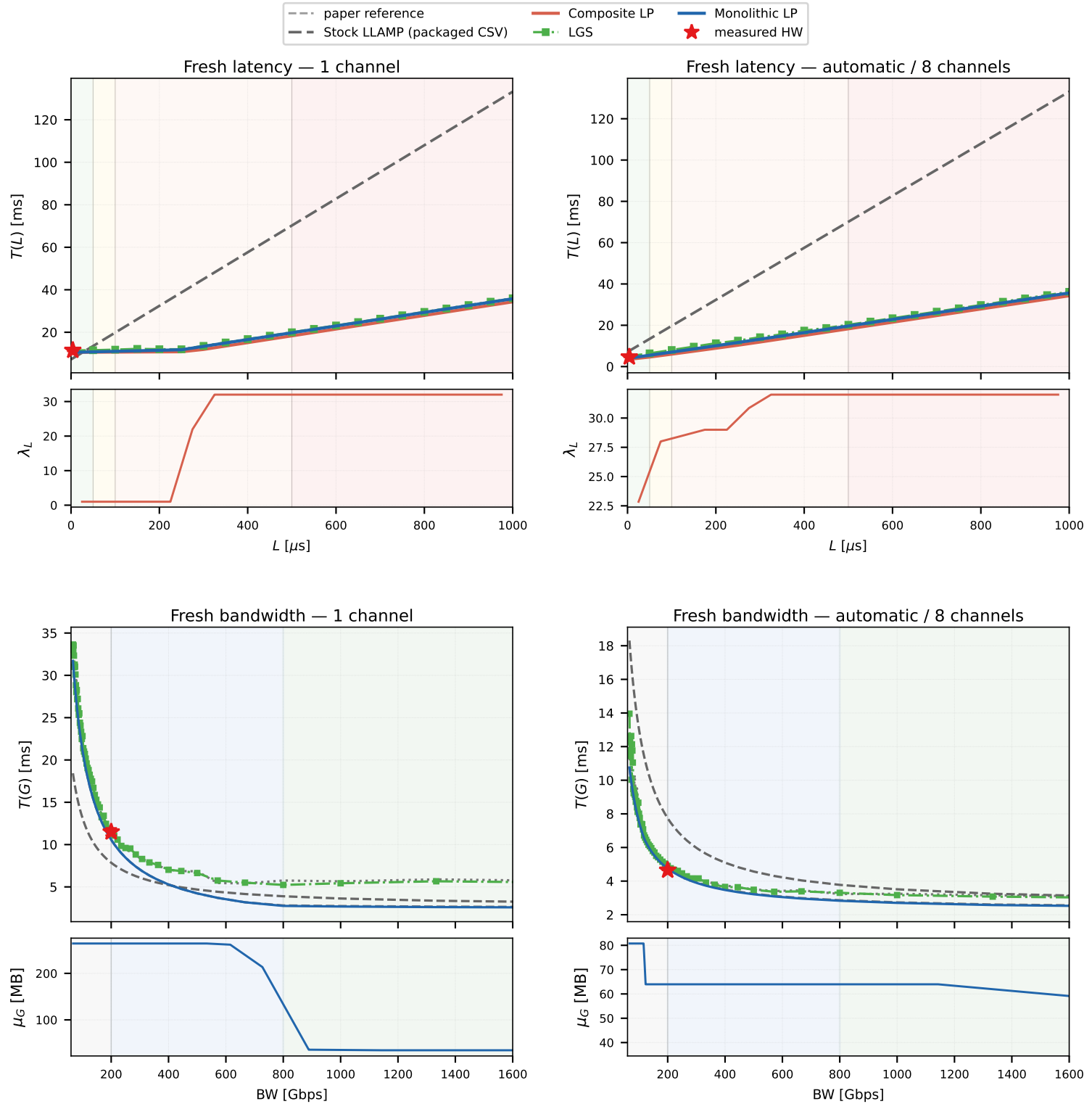
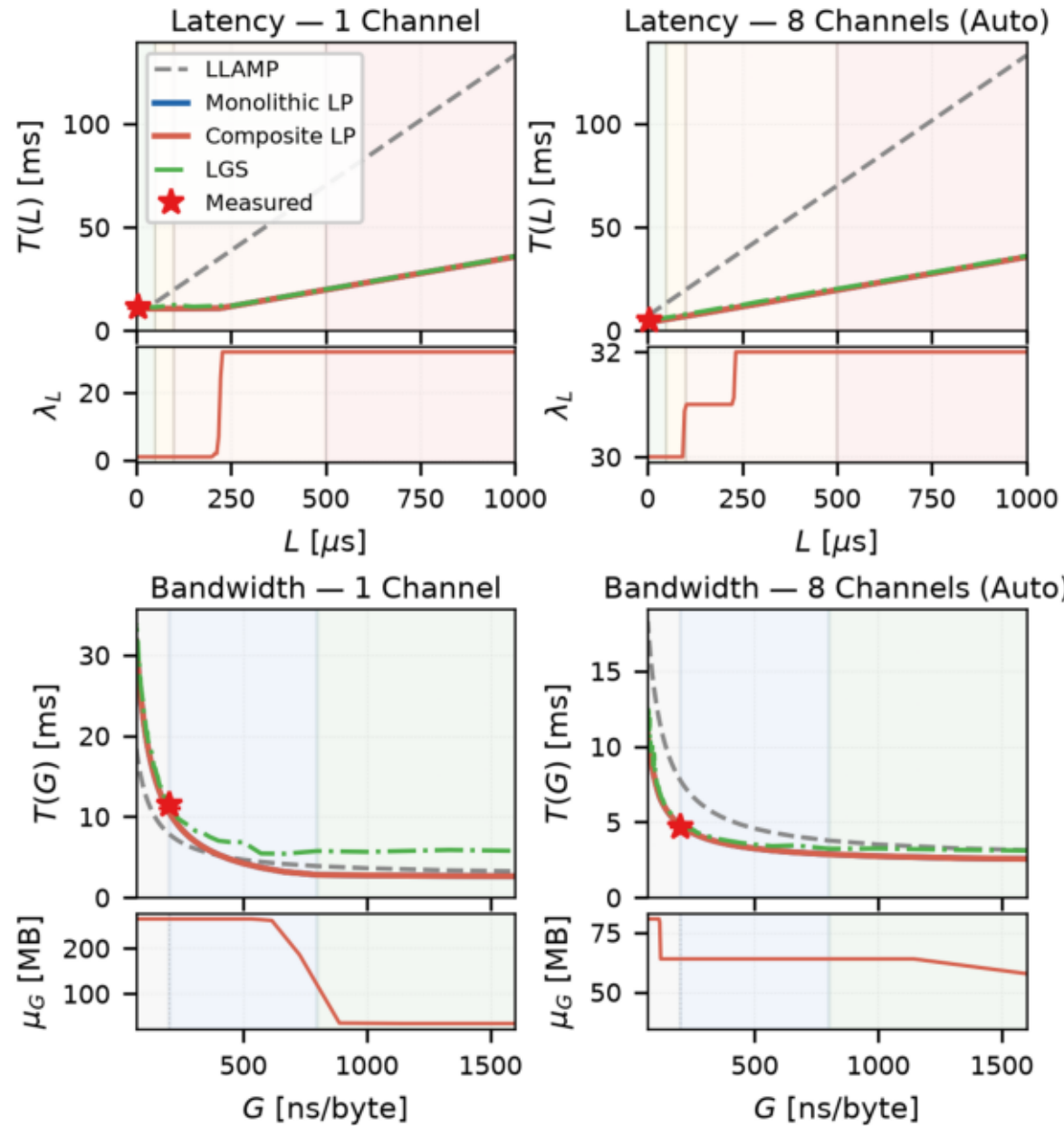
Left-hand plots on later pages are regenerated from the artifact's packaged paper data. They are references, not independent reruns. Right-hand plots combine outputs generated during this run with explicitly labelled packaged reference CSVs. Gray dashed curves are CSV-derived context; Stock LLAMP is included only in Figure 3, where it appears in the original paper, and is not a fresh solver run.

Figure 3 — 128 MiB AllReduce

Paper/reference on the left · fresh raw-input reproduction on the right

Paper/reference plot

AllReduce 128 MiB, 16 Nodes (64 GPUs)

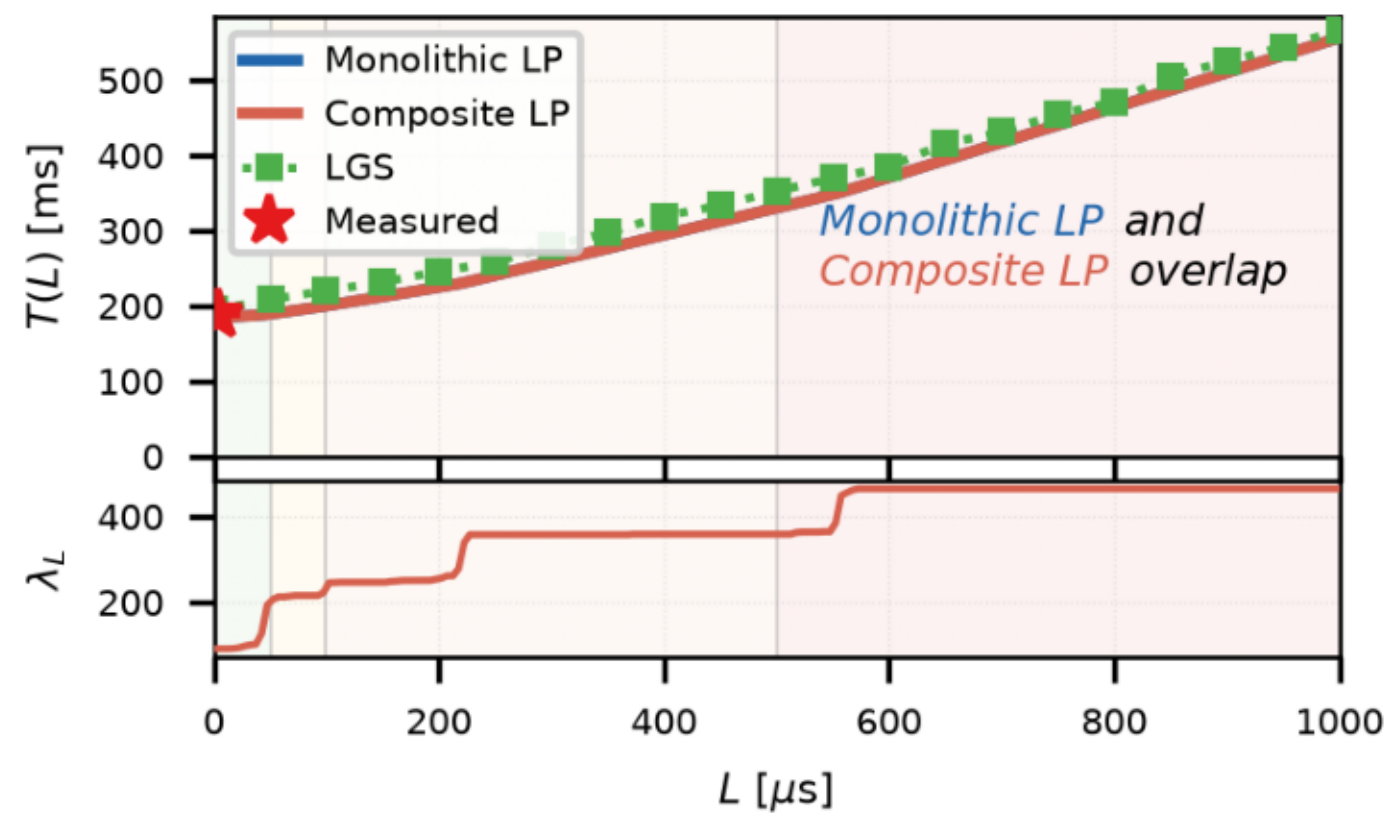


Evidence: 64 original NSYS reports per mode; 16 nodes, 64 ranks, Ring/Simple. The recovered historical generator establishes four physical NIC queues per GH200 node. Monolithic bandwidth: ch1 0.230% mean error; auto 0.523% mean error. Right-hand solid/marker curves are fresh; Stock LLAMP and paper-reference dashed curves are parsed from packaged CSVs.

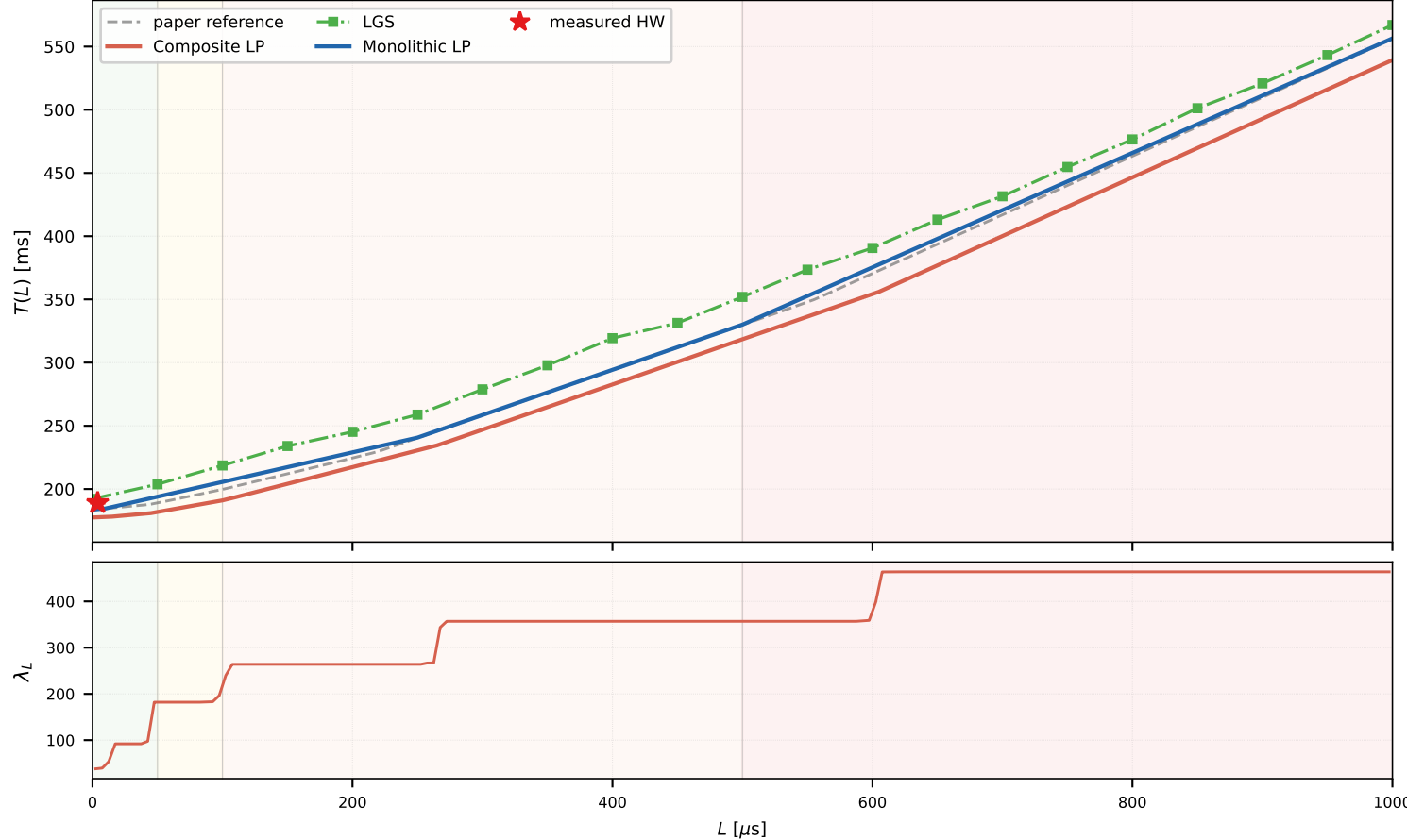
Figure 4 — mixed collectives

Paper/reference on the left · fresh end-to-end NSYS reproduction on the right

Paper/reference plot



Fresh end-to-end reproduction — job 1808340



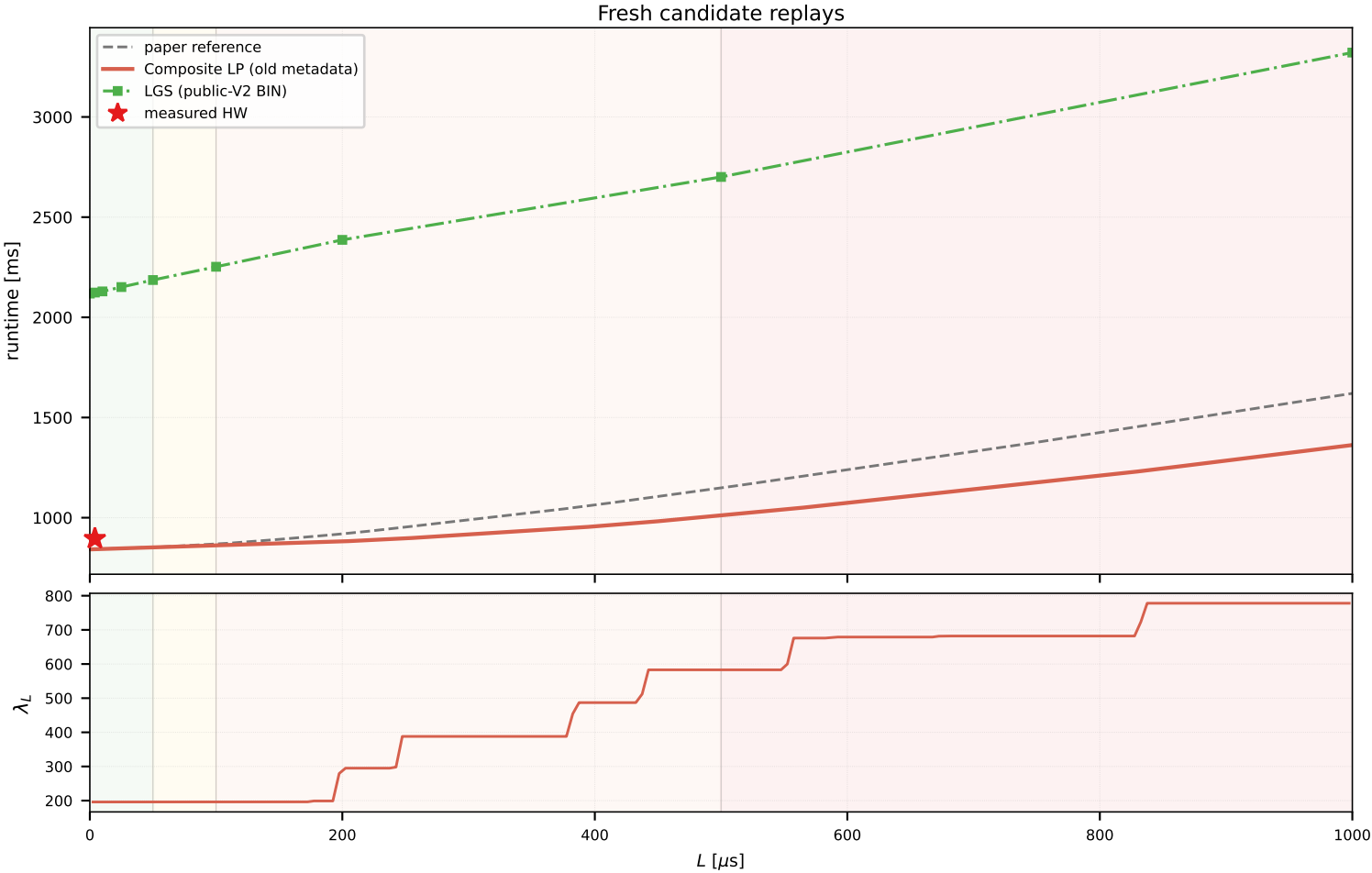
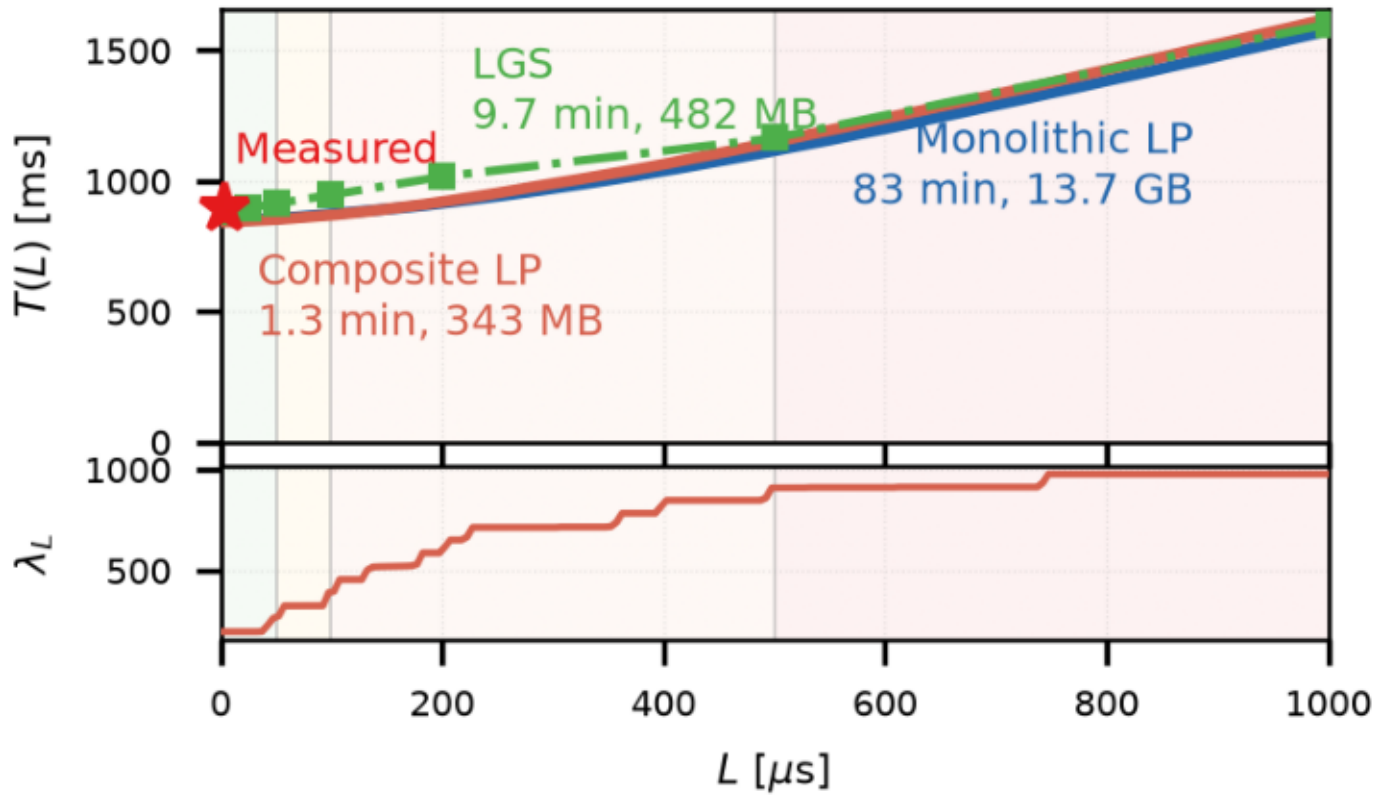
Provenance check (kept separate)

Job 1808340 is fully end-to-end from 64 recovered NSYS reports and is the numerical curve source (Monolithic max error 0.118%). Its 0.5–256 MiB messages conflict with the paper text. Corrected job 1791883 matches the written 16–64 MiB/seed 20262014 configuration but is $\sim 3\times$ slower and does not produce the published curve.

Figure 5 — Llama training iteration

Paper/reference on the left · recovered candidates on the right; exact paper input is still missing

Paper/reference plot



Exact paper input: unresolved
Both candidates identify Llama 7B, N4/GPU16, DP16
Paper label: Llama 8B
Raw one-iteration NSYS/GOAL: missing
Old metadata matches an older dev curve, not the paper slope
Public-V2 BIN is a different 16-rank program

The fresh old-metadata Composite starts at the same baseline as the paper but differs by 10.038% mean over the full curve. The public-V2 static BIN runs at roughly 2.1–3.3 s and is related provenance only. Neither candidate is promoted to exact. Stock LLAMP is omitted, matching the original Figure 5.

Figure 6 — non-Grok panels

Paper/reference on the left · fresh reproduction on the right; Grok excluded as requested

Paper/reference data

Fresh outputs

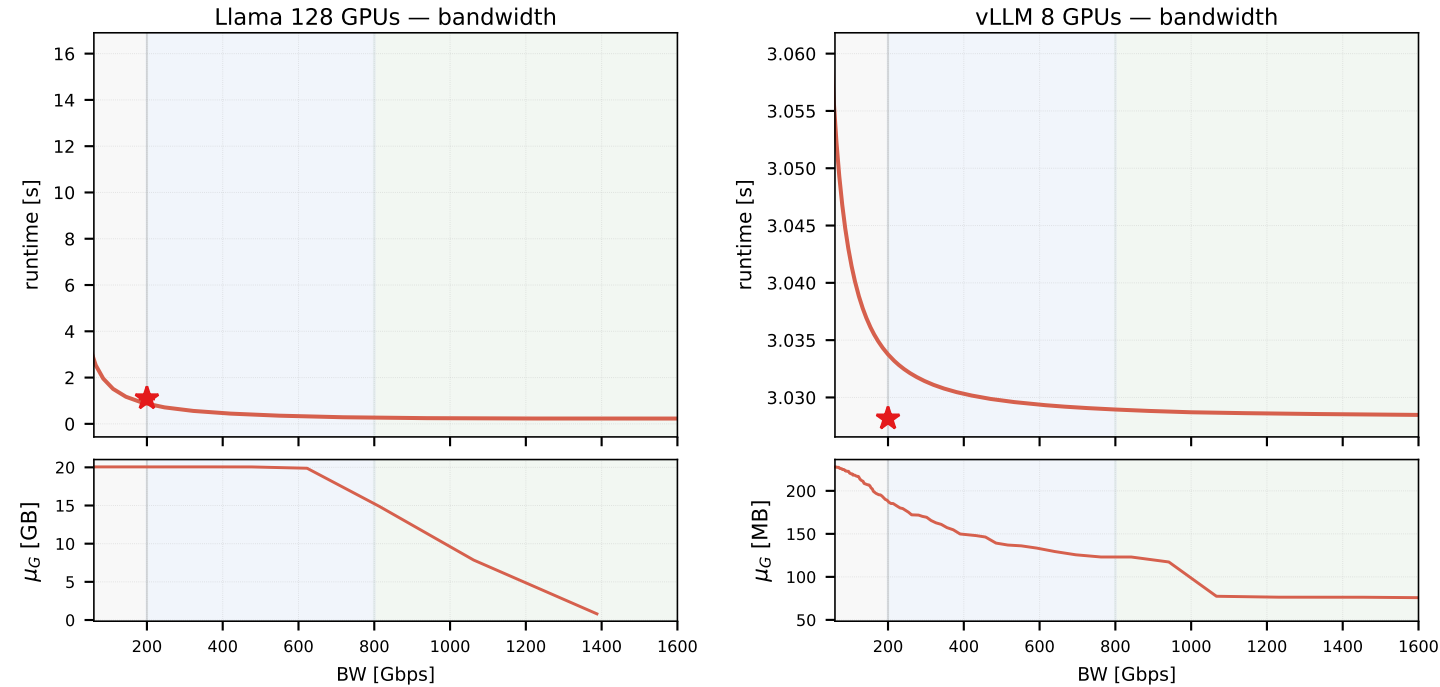
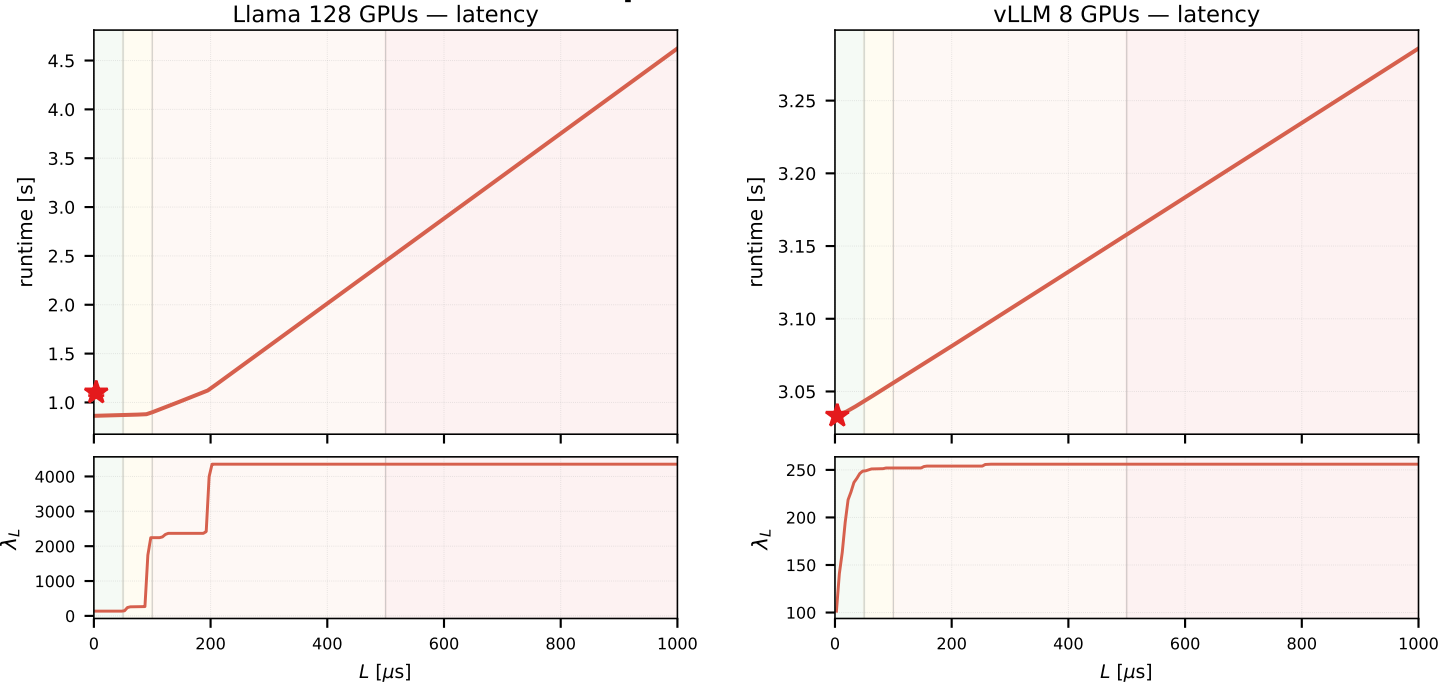
vLLM 8 GPUs — latency

BLOCKED

exact 8B/128-token NSYS and GOAL/BIN expired
70B trace deliberately not substituted

BLOCKED

exact 8B/128-token NSYS and GOAL/BIN expired
70B trace deliberately not substituted



Llama fresh regeneration starts from packaged NCCL metadata sidecars because no original N32 NSYS/GOAL/BIN survived locally. Every source identifies Llama 7B, while the paper labels 70B. The exact vLLM 8B job is known (1812656) but its original files expired.